

**Question:**  
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?  
Answer Choices:  
A.  $2.48 \times 10^{-8}$  cm  
B.  $5.92 \times 10^7$  cm/sec  
C.  $8.66 \times 10^{-6}$  cm  
D.  $3.14 \times 10^{-5}$  cm  
E.  $9.81 \times 10^{-4}$  cm  
F.  $5.27 \times 10^{-9}$  m  
G.  $1.23 \times 10^{-7}$  cm  
H.  $4.14 \times 10^{-10}$  m  
I.  $6.625 \times 10^{-27}$  erg sec  
J.  $0.512 \times 10^6$ eV

**Reasoning + Answer:**

</tool\_call> reasoning process here  
The de Broglie wavelength  $\lambda$  of a particle is given by :

$$\lambda = h / p$$

where :  
-  $h$  = Planck's constant =  $6.626 \times 10^{-34}$  J $\cdot$ s =  $6.626 \times 10^{-27}$  erg $\cdot$ s  
-  $p$  = momentum of the particle

For a non-relativistic electron , kinetic energy  $K = (1/2)mv^2$  , and momentum  $p = \sqrt{2mK}$

We can substitute to get :

$$\lambda = h / \sqrt{2mK}$$

We need to calculate  $\lambda$  for  $K = 1$  eV =  $1.602 \times 10^{-19}$  J

The mass of an electron  $m = 9.109 \times 10^{-31}$  kg

Now calculate  $p$  :

$$\begin{aligned} p &= \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}} \\ &= \sqrt{2.918 \times 10^{-50}} \\ &\approx 5.402 \times 10^{-25} \text{ kg}\cdot\text{m/s} \end{aligned}$$

Now calculate  $\lambda$  :

$$\begin{aligned} \lambda &= (6.626 \times 10^{-34}) / (5.402 \times 10^{-25}) \\ &\approx 1.227 \times 10^{-9} \text{ m} \end{aligned}$$

Now convert to cm :  
 $1 \text{ m} = 100 \text{ cm} \implies 1.227 \times 10^{-9} \text{ m} = 1.227 \times 10^{-7} \text{ cm}$

Now check the options in cm :

- A.  $2.48 \times 10^{-8}$  cm  $\implies$  too small
- B.  $5.92 \times 10^7$  cm/sec  $\implies$  not a wavelength
- C.  $8.66 \times 10^{-6}$  cm  $\implies$  too large
- D.  $3.14 \times 10^{-5}$  cm  $\implies$  too large
- E.  $9.81 \times 10^{-4}$  cm  $\implies$  too large
- F.  $5.27 \times 10^{-9}$  m  $\implies$  is  $5.27 \times 10^{-11}$  cm  $\implies$  not matching
- G.  $1.23 \times 10^{-7}$  cm  $\implies$  matches  $1.227 \times 10^{-7}$  cm
- H.  $4.14 \times 10^{-10}$  m  $\implies$   $4.14 \times 10^{-12}$  cm  $\implies$  no
- I.  $6.625 \times 10^{-27}$  erg sec  $\implies$  not a wavelength
- J.  $0.512 \times 10^6$  eV  $\implies$  energy , not wavelength

So , the correct value is approximately  $1.23 \times 10^{-7}$  cm

< answer > G </answer >